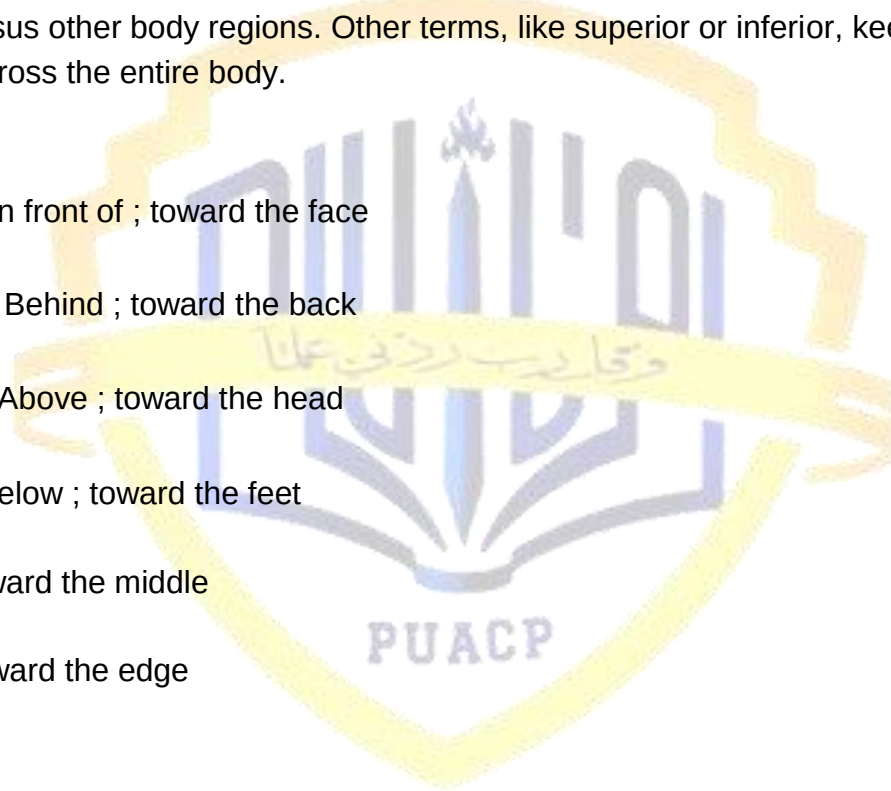


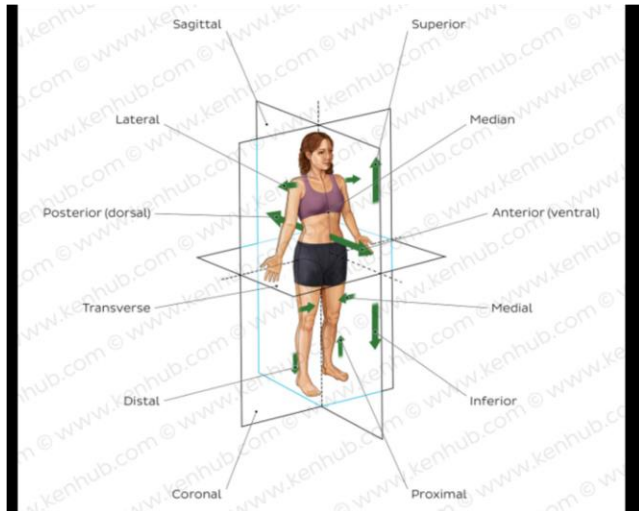
Neuro Anatomical Directional Terms And Planes Of References:-

Directional Terms:-

Directional terms are used to locate one structure, usually in relation to another structure. Some terms, like dorsal or ventral, are relative to the axis of the central nervous system, so the direction these terms define changes if used for brain regions versus other body regions. Other terms, like superior or inferior, keep their meaning across the entire body.

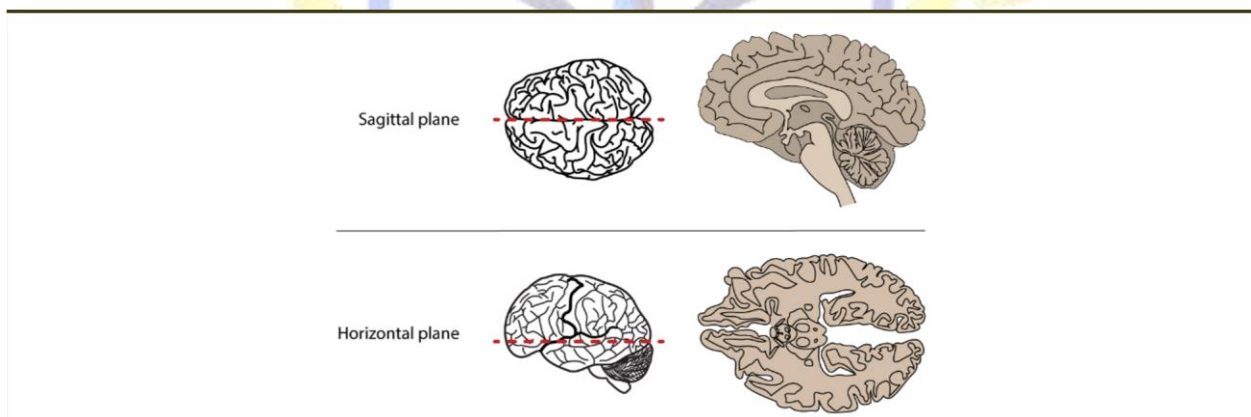
- Anterior: In front of ; toward the face
- Posterior: Behind ; toward the back
- Superior: Above ; toward the head
- Inferior: Below ; toward the feet
- Medial: toward the middle
- Lateral: toward the edge



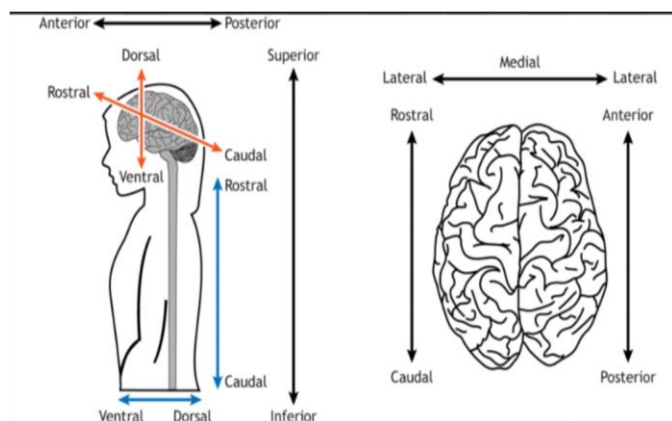


Planes of references:

- Sagittal Plane: Divides the brain into the left and right halves.
- Coronal Plane: Divides the brain into front (anterior) and back (posterior) portions
- Horizontal Plane: Divides the brain into top (anterior) and bottom (inferior) portions.



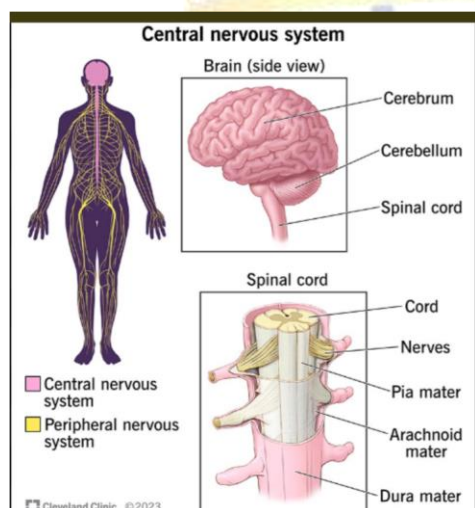
These directional terms and planes of reference are essential for describing the location of specific structures within the nervous system, as well as for understanding the relationships between different regions of the brain and spinal cord.



Directional terms used to locate nervous system structures. The dorsal / ventral and rostral / caudal pairs point in different directions depending on if they are referring to the axis of the brain (orange arrows) or the axis of the spinal cord (blue arrows).

Structures of Nervous System, their functions and connections:-

The nervous system consists of two main parts: the central nervous system (CNS) and the peripheral nervous system (PNS). Here are the functions and structures.

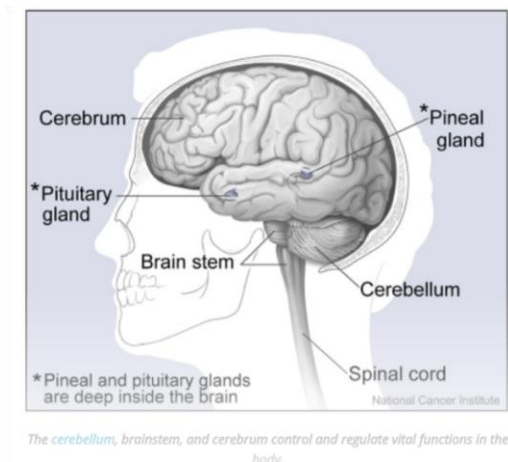


1- Central nervous system (CNS):- The CNS consists of the brain and spinal cord, which are enclosed within the skull and vertebral column.

Brain:- The brain is the control center of the nervous system and is responsible for receiving, processing, and integrating sensory information, initiating motor commands,

and regulating homeostasis. It is divided into several regions, including the cerebellum, brainstem, and diencephalon.

Spinal Cord:- The spinal cord is a long, thin structure that extends from the brainstem down through the vertebral column. It is responsible for transmitting signals between the brain and the rest of the body, as well as for integrating some reflexes.



Functional Divisions of the Nervous System:-

All of the hormones and endocrine systems discussed so far are subject to some degree of direct or indirect regulation by the nervous system, particularly the autonomic nervous system. Before discussing the neural regulation of endocrine systems, it is important to have a rudimentary understanding of the functional divisions of the nervous system.

2- Peripheral nervous system (PNS):- The PNS consists of all the nerves and ganglia outside of the CNS.

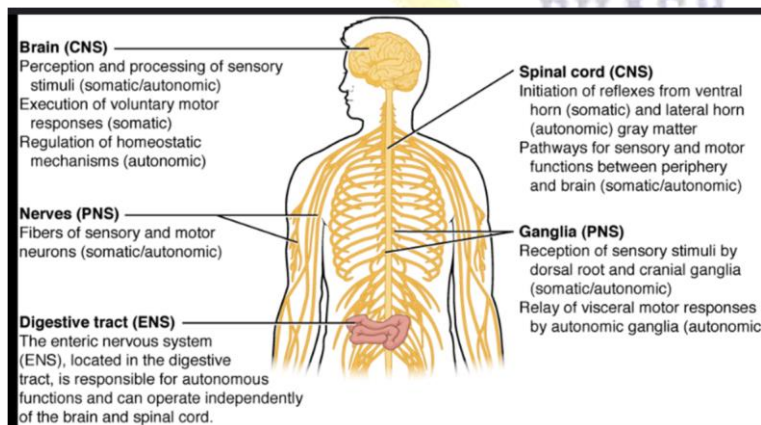
Cranial nerves:- There are 12 pairs of cranial nerves that originate from the brainstem and innervate structures in the head and neck, such as the eyes, ears, and tongue.

Spinal nerves:- There are 31 pairs of spinal nerves that emerge from the spinal cord and innervate structures in the rest of the body, such as the limbs, trunks, and internal organs.

Autonomic nervous system (ANS):- The autonomic nervous system is a division of the PNS that controls involuntary processes such as heart rate, breathing, digestion, and glandular secretion. It is further divided into the sympathetic and parasympathetic nervous systems.

Enteric nervous system(ENS) :- The enteric nervous system is a network of neurons and ganglia that is embedded in the wall of the gastrointestinal tract. It is responsible for regulating digestion and other gastrointestinal functions.

The nervous system probably includes the **brain**, the nervous tissues contained within the cranium, and the **spinal cord**, the extension of nervous tissue within the vertebral column. Additionally, the nervous tissue that reaches out from the brain and spinal cord to the rest of the body (**nerves**) are also part of the nervous system. We can anatomically divide the nervous system into two major regions; the **central nervous system (CNS)** is the brain and spinal cord, the **peripheral nervous system (PNS)** is the nerves. The brain is contained within the cranial cavity of the skull, and the spinal cord is contained within the vertebral canal of the vertebral column. The peripheral nervous system is so named because it is in the periphery meaning beyond the brain and spinal cord. The PNS is responsible for transmitting signals between the CNS and the rest of the body. The ANS and ENS play a crucial role in regulating physiological processes such as digestion, metabolism, and immune function.



Functional Divisions of the Nervous System:-

The nervous system can also be divided on the basis of its functions. The nervous system is involved in receiving information about the environment around us (sensory functions, **sensation**) and generating responses to that information (motor functions, **responses**) and coordinating the two (**integration**) .

Sensation refers to receiving information about the environment, either what is happening outside (i.e; heat from the sun) or inside the body (i.e; heat from muscle activity). These sensations are known as **stimuli** (singular = **stimulus**) and different sensory receptors are responsible for detecting different stimuli. Sensory information travels towards the CNS through the PNS. When information arises from sensory receptors in the skin, skeletal muscles, or joints, it is transmitted to the CNS using **somatic sensory** neurons; when information arises from sensory receptors in the blood vessels or internal organs, it is transmitted to the CNS using **visceral sensory** neurons.

Response is the nervous system that produces a response in **effector organs** (such as muscles or glands) due to the sensory stimuli. The motor (**efferent**) branch of the PNS carries signals away from the CNS to the effector organs. When the effector organ is a skeletal muscle, the neuron carrying the information is called a **somatic motor** neuron; when the effector organ is cardiac or smooth muscles or glandular tissue, the neuron carrying the information is called an **autonomic motor** neuron. Voluntary responses are governed by somatic motor neurons and involuntary responses are governed by the autonomic motor neurons.

Integration: Stimuli that are detected by sensory structures are communicated to the nervous system where information is processed. In the CNS, information from some stimuli is compared with, or integrated with, information from other stimuli or memories of previous stimuli. Then, a motor neuron is activated to initiate a response from the effector organ. This process during which sensory information is processed and a motor response generated is called **integration**.

The nervous system is functionally divided into two main divisions: the somatic nervous system (SNS) and the autonomic nervous system (ANS).

The **peripheral nervous system (PNS)** contains neurons outside of the brain and spinal cord and consists of the **somatic nervous system (SNS)**, the **autonomic nervous**

system (ANS), and the **enteric nervous system (ENS)**. While CNS neurons cannot be regenerated if damaged, neurons of the PNS can recover, though they often fail to reform connections made prior to damage.

1-The somatic nervous system, under conscious control, contains sensory neurons bringing information to the CNS, and motor neurons carrying commands from the CNS. The ANS controls almost every other tissue in the body, including cardiac muscles, smooth muscles, organs, etc.

Motor neurons: These are different neurons that transmit signals from the CNS to the skeletal muscles, causing them to contract and produce movement.

Sensory neurons: These are afferent neurons that transmit signals from sensory receptors in the skin, muscles, joints to the CNS, allowing us to perceive and respond to external stimuli.

2- Autonomic Nervous System (ANS), the ANS controls involuntary physiological processes, including heart rate, breathing, digestion, and glandular secretion. It is divided into two opposing systems that continually monitor and antagonistically control physiological systems. The two components of the autonomic nervous system are the sympathetic **nervous system (SNS)** and the **parasympathetic nervous system (PNS)**.

Sympathetic Nervous System: The sympathetic nervous system prepares the body for action by increasing heart rate, dilating airways, and increasing blood flow to the muscles. It is often called the “fight or flight” response because it prepares the body to respond to a threat.

Parasympathetic Nervous System: The parasympathetic nervous system slows down physiological processes and promotes relaxation and digestion. It is often called the “rest and digest” response because it promotes activities that occur during times of rest and recovery.

These functional divisions of the nervous system are crucial for maintaining homeostasis, or the stable internal environment of the body. The SNS allows us to interact with the external environment by responding to sensory stimuli and producing voluntary movements. The ANS controls involuntary processes that are necessary for survival, such as heart rate and digestion, allowing us to respond to changes in our internal and external environment.